EC-321 Microprocessor Engineering

End-Term Examination, Odd Semester 2025-26, IIT (BHU) Varanasi

Date & Time: 18-Nov-2025 09.00 am to 12.00 pm

Max. Duration: 3 Hours **Max. Marks : 100**

General Instructions:

i. Answer ***ALL*** the questions; Plan for a proper "***time-management***", before starting to write the exam

ii. State valid logical reasons, whenever making any relevant additional assumptions in your answers

iii. ***Every successive answer MUST BE started in a fresh page; or else draw a line between two answers***

SECTION - A (15 x 3 = 45 Marks)

(Each question in this section carries 3 Marks)

1. Please elaborate on any three differences between "*Classical Computing*" and "*Quantum Computing*".
2. Give the bit-structure of "Program Status Word" (PSW) register of 8051-Microcontroller. Explain the significance of every bit present in this register.
3. What is the significance of using "Motor Driver ICs", while interfacing Microprocessors & Microcontrollers with Stepper motors. Please name any one Motor Driver IC that is familiar to you.
4. Draw a block diagram to represent the various signals involved in "Microprocessor mediated Bus Interfacing". Explain the functions of every signal involved in Microprocessor mediated Bus Interfacing, along with example of a bus timing-diagram.
5. What is the significance of using "Cache Memories" in computing systems? Draw a block diagram to represent Cache memory interfacing with Microprocessors. Define the terms "Cache Hit" & "Cache Miss".
6. Explain the roles of following three signals seen in the pin-diagram of 8086 Microprocessor: (i). NMI (ii). BHE (iii). MN/$\overline{MX}$
7. Give full-form expansions of these terms: (i). SFR (ii). MIPS (iii). HDL (iv). DRAM (v). MMU (vi). SPICE
8. What are the advantages of using "Multi-core Processors"? Can you suggest any two key differences between Single-core Processors and Multi-core Processors ?
9. Draw the hardware block diagram of 8155 chip. In the diagram, mark all the important signals present in 8155 chip that are useful in interfacing to the external world *(Only diagram is sufficient; No explanations are required)*
10. Assume that you are given the task of interfacing a 4x4 keypad matrix with a 8085 Microprocessor, through a Programmable Peripheral Interface Chip. Can you show a circuit-diagram representation to achieve this task. Also, brief upon any technique that can be used to implement "Key-Debouncing" feature in this system.
11. Explain the working of SIM instruction in 8085 Microprocessor, along with depiction of associated 8-bit pattern for the Accumulator.
12. What do you understand by the terms "Immediate Addressing mode" and "Implicit Addressing Mode" ? Explain these two types of addressing modes, with one example each.
13. Explain the working of "ROR" and "RCL" instructions in 8086 microprocessor, along with appropriate diagrams to depict the process of bit-shifting in each of these two instructions.
14. With regards to "Qubits" used in Quantum Computers, what do you understand by the terms "Superposition" and "Entanglement" ?
15. In your own words, define the term "*Humanoid Robot*". In an intuitive way, briefly describe as to how Microprocessors are useful in the area of Robotic System Design.

SECTION - B (1 x 5 = 5 Marks)

(Each question in this section carries 5 Marks)

16. Briefly explain the working of following 8086 instructions along with an example each: (i). JAE (ii). CLD (iii). NEG (iv). SBB (v). CMP

SECTION - C (1 x 10 = 10 Marks)

(Each question in this section carries 10 Marks)

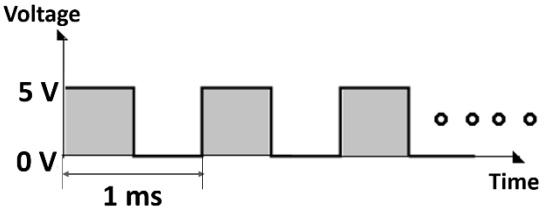
17. (i). Draw the internal block diagram of 8051 Microcontroller, depicting all important hardware modules interfaced to the processor-core present in 8051-chip, at a gross level. (ii). Draw the Micro-architectural block diagram of 8051 Microcontroller, and also explain the salient features of the architecture.

-P.T.O.

SECTION – D (Problem-Solving)

(Marks for every question in this section is indicated across each question)

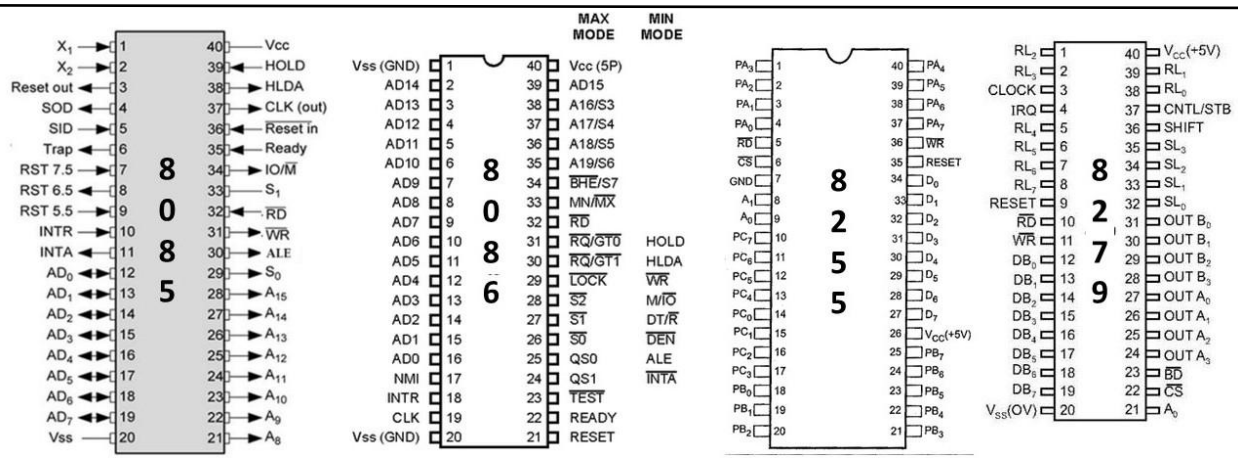
18. Write an Assembly Language Program for 8085 Microprocessor to generate a continuous square wave pattern as shown in the figure given here. The square waveform is to be sent as output on the pin D0 (LSB of the data bus) of the Microprocessor, and has to be further driven onto an output port with address 8A-H. Assume that the processor works with a clock frequency of ***3.08 MHz***. [*Note:* For every instruction present in your ALP, you may assume arbitrary values of **T**-states within the range *(4-T, 14-T)*, for performing delay calculations. In your answer sheet, you shall write the program as a 4-column format, namely: i. Memory location range ii. Mnemonics (program code) iii. Comments for each instruction, iv. T-state value assumed for the instruction. In addition to this, you must also clearly indicate the delay calculations, below the table]. ***{10 Marks}***



19. Assume that you are required to interface 2 numbers of ***8K X 8*** SRAM ICs, and 2 numbers of ***4K X 8*** EPROM ICs with a 8086 Microprocessor. The memory system is to be designed in such a way that the RAM and ROM memories are able to transfer data values onto the Microprocessor bus, both in byte-mode and word-mode. As a designer, you have been given the freedom to choose the ranges of memory addresses for RAM and ROM. Design the *processor-memory interface* by satisfying all the given constraints. In your answer sheet, give your design result, in terms of a detailed circuit diagram with associated signals clearly labeled in it. The circuit diagram should also clearly indicate all the logic-circuits required to properly enable the chip-selects for all the 4 memory ICs present in the system. In addition, also show a table indicating the range of memory addresses chosen by you, for the RAM and ROM ICs, where the address locations are to be indicated both in binary format and hexadecimal format. ***{12 Marks}***

20. Design a hardware system using 8279 chip, to interface an ***8 x 8*** key-pad with a 8085 Microprocessor. In the same system, you are also required to interface 4 numbers of 7-segment displays (common-cathode based) for output purposes. Give your design result in terms of a detailed circuit diagram, showing all required hardware components, and their interconnections. In your answer sheet, also briefly explain the working of the hardware system, in your own words, below the circuit diagram. ***{12 Marks}***

21. Assume that, using an 8086 Microprocessor, you are required to design a *control-panel* for an *Elevator (Lift)* situated in a multi-storey building with 10-floors. The hardware design is governed by fulfilling the following specifications: ***i).*** The microprocessor is required to receive inputs from a user regarding floor selection, door open/close, etc., through input buttons from a keypad-matrix ***ii).*** The microprocessor shall interface with a high-power DC-motor situated on the top of the building, based on the inputs chosen by the user, so as to transport the lift to the required floor, ***iii).*** The lift-panel must contain two 7-segment displays to display the instantaneous floor-position (as from 00 to 10). ***iv).*** The control-panel must also talk to a flame-sensor present inside the lift, to detect any unfortunate event of a fire inside the lift ***v).*** Upon detection of fire, the microprocessor must trigger an alarm-bell situated in ground floor; *For this design problem, give your design-solution, in terms of following: i) Block-diagram for the hardware-design, ii). Flowchart-diagram of algorithm that can be used to program the 8086-processor to satisfy all given design specifications.* ***{6 Marks}***



--Pin-outs of some important ICs taught in this course--